2.6 CIRCUIT BREAKER

A circuit breaker is an automobile operated electrical switch designed to protect an electrical circuit from damage caused due to excess current, typically resulting from an over-load or short-circuit. Its basic function is to stop current flow after short-circuit or over-load is detected. There are different types of circuit breakers and are summarized below:

2.6.1 MCB (Miniature Circuit Breaker)

Presently, we use MCB in low-voltage electrical network instead of fuse. MCB is an electromechanical device which protects an electrical circuit from over-current. An MCB is better alternative to fuse, since it does not require replacement once an over-load is detected. Unlike fuse, an MCB can easily reset and thus offers improved operational safety.

Working principle

MCBs are either electromagnetic or a bi-metallic strip. In either case, when turned on, the breaker allows electrical current to pass from a bottom to an upper terminal across the solenoid or strip. When the current reaches un-safe levels, the magnetic force of the solenoid becomes so strong, that a metal lever within the switch mechanism is thrown and the current gets cut-off. Alternatively, the metal strip bends throwing the switch and disconnecting the connection. To reset the flow of electricity after the problem is resolved, the switch can simply be tuned back to on, thus reconnecting the circuit. Figure 2.38 shows a layout of MCB.



Figure 2.38 Layout of Miniature Circuit Breaker

Difference between Fuse and MCB

The difference between MCB and Fuse is summarized in Table 2.3.

Table 2.3: Difference between Fuse and MCB

Fuse	MCB
Fuse is usually made up of a fuse wire. It is an alloy which has a no melting point	MCB has a tripping circuit
Fuse works on the electrical and thermal properties of the conducting materials.	MCB works on the electromagnetism and switching principle
Fuses can be used only once	MCB can be used a number of times
Fuse cannot be used as an ON/OFF switch	MCB can be used as an ON/OFF switch
Operating time of Fuse is Very less (0.002 sec)	Operating time is comparatively more than that of fuse (0.02-0.05 sec)
Cost of fuse is low	Cost of MCB is high

ELEC_EQUIP_Chapter02.indd 54 11/10/2020 12:23:51 PM